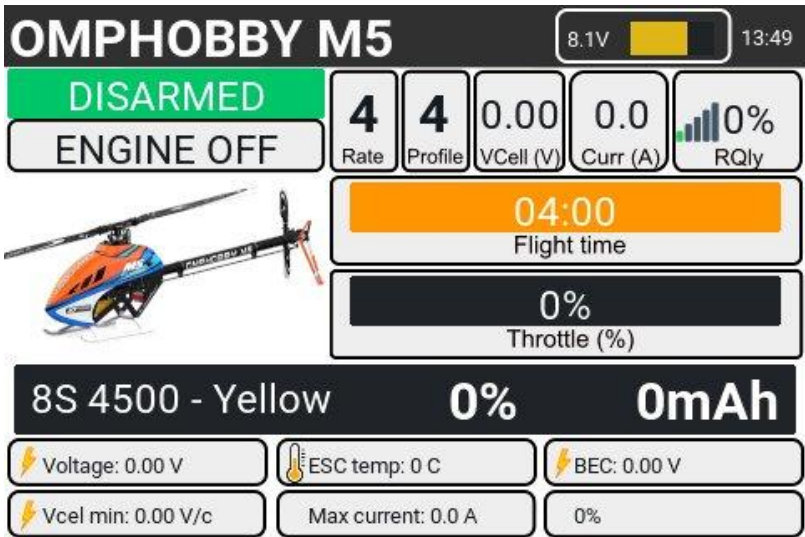


SSODB Battery Manager

User & Installation Manual



An EdgeTX widget for Rotorflight helicopters
September 2026

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1. Installation

1.1. What you need

- An EdgeTX transmitter
- A Rotorflight 2 flight controller with telemetry setup (see 1.4)
- The Rotorflight Lua scripts installed in /SCRIPTS/RF2/ on your SD card (these come with Rotorflight — the widget uses them to talk to the FC)

1.2. Copying the files

Everything lives in a single folder on the transmitter's SD card. **The folder must be named exactly SSODB** — some of the widget's internal paths depend on it.

/WIDGETS/SSODB/	
main.lua	the widget itself
pidtune.lua	the PID tuning screen
rf2util.lua	shared flight-controller communication
Background.png	the background artwork
Modelimage/	put your model photos here
M5.bmp	
M7R.bmp	
icons/	link-quality icons

1. Power off the transmitter and remove the SD card, or connect it by USB.
2. Copy all files to the /Widgets folder
3. Put the card back and power the transmitter on.

You do **not** need to create batteries.lua or model.lua by hand — the widget writes them itself the first time it needs to.

1.3. Adding the widget to a screen

1. Open the model's UI setup.
2. Choose a screen and set its layout to app mode — one zone.
3. Click on "Setup widget"
4. Tap the screen (or press enter) and Click "Select widget". Pick SSODB from the list.
5. Leave the screen setup. The widget appears.

The widget must be full screen. In a smaller zone it will just display "Set to fullscreen" and do nothing else.

IMPORTANT!: Before you can interact with the widget, you'll need to activate it / focus on it. To do this, long press the screen or enter button until the EdgeTx logo disappears.

1.4. Rotorflight telemetry setup

To get the most out of the widget, setup the telemetry sensors in the Receiver page in the Rotorflight Configurator as follows:

Telemetry Sensors		Sort	Select
Battery			4 / 6
Battery Voltage			☒
Battery Current			☐
Battery Consumption			☒
Battery Charge Level			☒
Battery Cell Count			☐
Battery Cell Voltage			☒
Voltage			1 / 4
ESC Voltage			☐
BEC Voltage			☒
Bus Voltage			☐
MCU Voltage			☐
Current			1 / 4
ESC Current			☒
BEC Current			☐
Bus Current			☐
MCU Current			☐
Temperature			1 / 3
ESC Temp			☒
BEC Temp			☐
MCU Temp			☐
ESC #1			0 / 12
ESC #2			0 / 6
RPM			1 / 2
Headspeed			☒
Tailspeed			☐
Barometer			0 / 2
Gyro			0 / 9
GPS			0 / 8
Status			0 / 7
Profile			3 / 4
PID Profile			☒
Rates Profile			☒
Battery Profile			☒
LED Profile			☐
Control			0 / 6
System			0 / 4
Debug			0 / 8

2. First-time setup

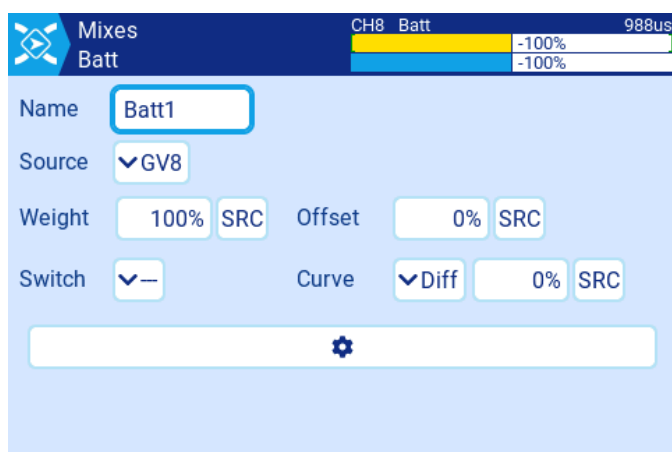
Before using the widget, do these three things.

2.1. Open the widget settings and check the sources.

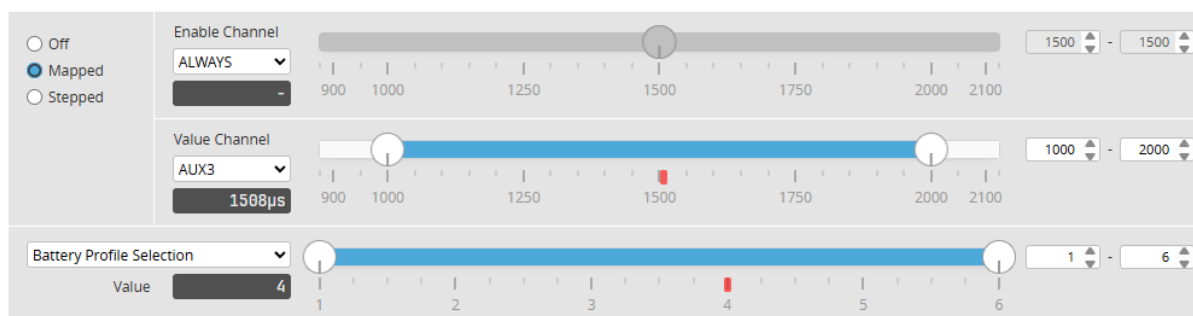
The widget settings can be accessed through the UI setup screen. Select the screen and click Setup widgets. Work through the list in section 3 (widget settings explained).

2.2. Set up the global variable.

The widget writes a value to one of the transmitter's global variables (GV8 by default), and Rotorflight uses that to choose its battery profile. In EdgeTX, setup a mix for the channel used to transmit the selected battery to Rotorflight.



In the Rotorflight configurator, in the adjustments tab, you can set the Battery profile selection to the channel you assigned in EdgeTx.



2.3. Add your batteries.

See section 7.

2.4. Add your model images

Copy your model images to /Widgets/SSODB/Modelimage

3. Widget settings explained



Setting	What it is	Default value
MinCell	Sensor holding the lowest cell voltage seen	Vcel-
MaxCurr	Sensor holding the highest current seen	Iesc+
MaxEscT	Sensor holding the highest ESC temperature seen	Tesc+
MinRQly	Sensor holding the worst link quality seen	RQly-
GV_Select	Which global variable to write the battery slot into	8
TimerSrc	Which of the model's three timers to display	1



Setting	What it is	Typical value
ArmSrc	Your arming switch or the ARM logical switch	ARM
ModeSrc	Your engine / throttle-hold switch	e.g. SF
RpmSrc	Headspeed telemetry source, if the sensor isn't auto-detected	Hspd
RateSrc	Source that selects the rate bank	Bank
NumRates	How many rate profiles you use	4

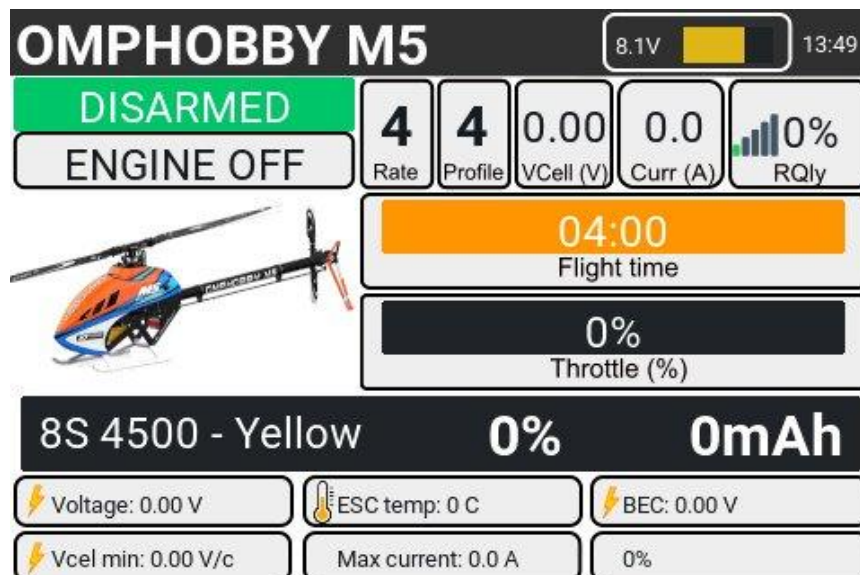


Setting	What it is	Typical value
ThrSrc	Throttle percentage telemetry sensor	THR
ProfSrc	Source that selects the PID profile	Bank
NumProfs	How many PID profiles you use	4
BkGround	Background colour behind the cards	white

About the sensor settings

Most telemetry values are found **automatically** — the widget knows the common Rotorflight sensor names and tries them in turn. The settings above are there for when your sensor is named something unusual, or when a value comes from a switch rather than a sensor.

4. The flight screen



This is the main display it contains:

Top row — the model name (tap it to open PID tuning), the transmitter battery voltage, and the clock.

Status cards — armed state, engine state, the current rate bank and PID profile, cell voltage, current, and link quality.

Model photo — tap it to choose a different photo.

Flight timer — an orange bar that empties as the timer counts down, with the time shown inside it.

Throttle — a bar showing throttle percentage, updated about once a second.

Battery bar — the selected pack's name, the percentage remaining, and the mAh consumed. It turns orange below 80% and red below 20%. Tap it to go back to the battery select screen.

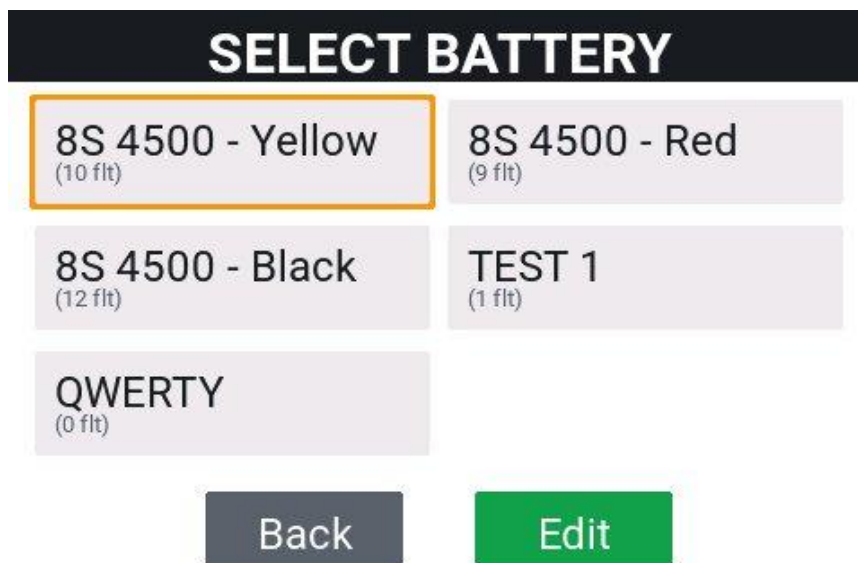
Bottom row — pack voltage, ESC temperature, BEC voltage, lowest cell voltage of the flight, highest current of the flight, and worst link quality.

The engine card

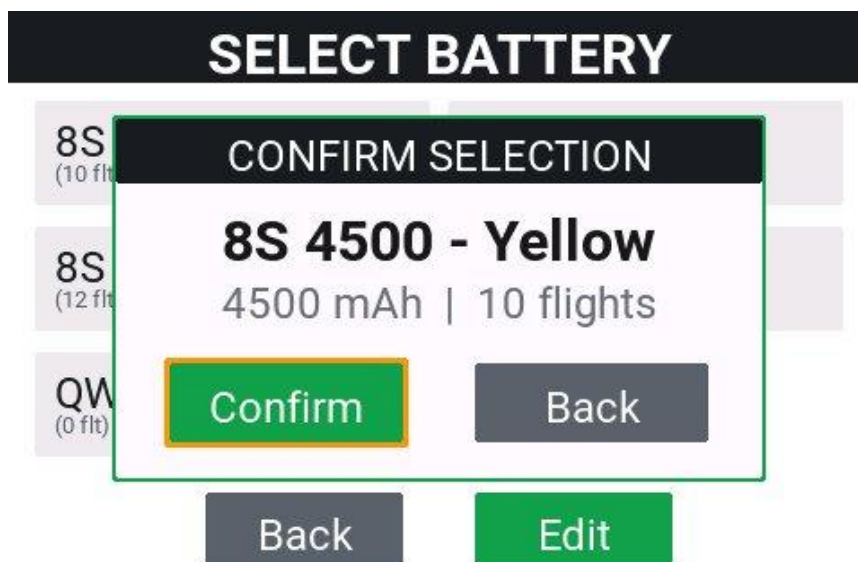
It says	It means
ENGINE OFF	Your engine switch is off.
IDLE	Your engine switch is in its middle position.
ENGINE ON (orange)	Engine commanded on, but the head is turning below 100 rpm.
1850 RPM	Engine commanded on and the head is turning.

5. Choosing a battery

When a model connects, the widget automatically jumps to this screen. You can also access this screen by tapping the battery bar on the flight screen.



Each pack shows its name and how many flights it has done. Tap one and you get a confirmation:



Tap Confirm and three things happen:

1. The global variable is set, so Rotorflight switches to that battery profile.
2. The pack's usable capacity and cell-voltage limit are sent to the flight controller.
3. The widget returns to the flight screen and starts timing.

If a small orange "FC sync..." note appears at the top right, the widget is still talking to the flight controller. It clears by itself. If it turns red with an error, see section 14.

Only the batteries assigned to the connected model are listed. If none of your packs are assigned to that model, all of them are shown.

The Edit button takes you to the battery list.

6. Managing your batteries



Each row shows the pack's name, its capacity, its usable percentage, and HV if it is a high-voltage pack. The green bar on the left edge marks it as a stored pack.

- Tap a row to edit that pack.
- Tap the red X to delete it.
- “+ Add Battery” creates a new pack and opens it for editing.
- Use the ^ and v buttons on the right when you have more packs than fit on the screen.

When deleting a battery, the widget asks for a confirmation first — When confirmed, deletions are written to the card straight away and cannot be undone:



7. Editing a pack

< List

Saved!

Save

Name

Batt 9 >

Capacity

2200 mAh >

Usable

100% (2200 mAh) >

Cell type

LiPo 4.20 V/cell

Flights

0 >

Models

Global (all models) >

Row	What it does
Name	The name of the battery pack
Capacity	The pack's rated capacity in mAh, 0–20000.
Usable	What percentage of the rated capacity you actually want to use, 1–100%. The figure in brackets is what will be sent to the flight controller.
Cell type	Tap to toggle between LiPo (4.20 V/cell) and HV LiPo (4.35 V/cell).
Flights	The flight counter. You can edit it, for example when adding a pack you have already been flying.
Models	Which models this pack is assigned to. See section 8.

Tap Save when you are done. A green "Saved!" appears in the header:

Saved!

Back

12S 5100 RED

5100mAh 75%

X

12S 5500 YELLOW

5500mAh 75%

X

HV 12S 6500

6500mAh 75% HV

X

TEST 1

10mAh 100%

X

QWERTY

2200mAh 75%

X

+ Add Battery

7.1. About "Usable"

This is the safety margin. If you have a 5100 mAh pack and set Usable to 75%, the widget tells the flight controller the pack holds 3825 mAh. Rotorflight's percentage readout and low-capacity warnings are then based on that smaller figure, so you land with a reserve rather than running the pack flat.

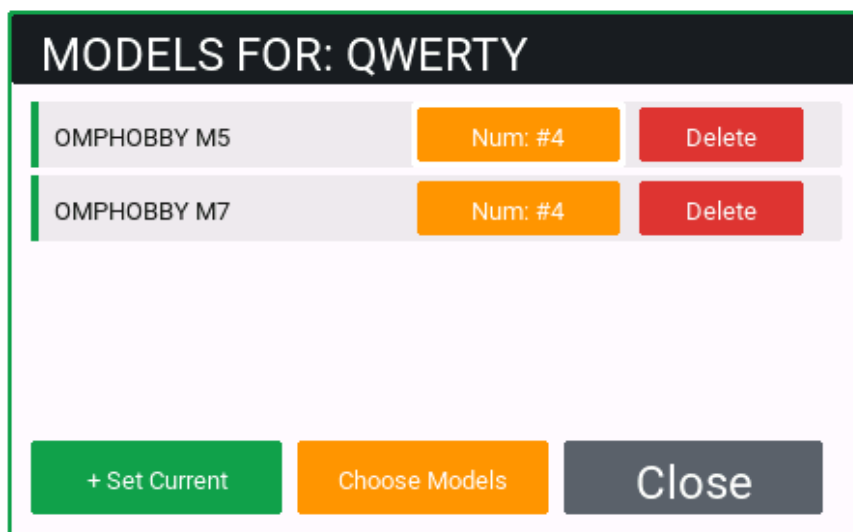
7.2. About "Cell type"

This decides the maximum cell voltage sent to the flight controller: 4.30 V per cell for standard LiPo, 4.40 V per cell for HV. These are ceilings set a little above each chemistry's full-charge voltage so a freshly charged pack does not trip the limit.

8. Assigning batteries to models

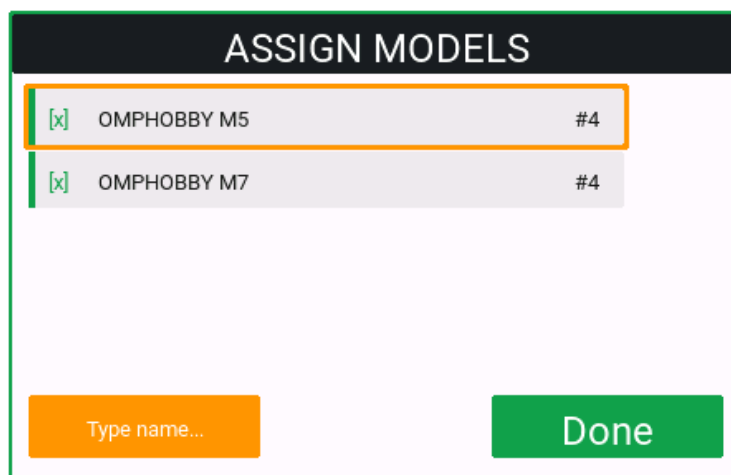
Assigning a pack to a model does two things: the pack only appears in the select list for that model, and it gets a **number from 1 to 6** which decides the Rotorflight battery profile used.

From the battery editor, tap Models. The following screen appears:



Functions:

- “+ Set Current” assigns the pack to the model currently connected.
- “Choose Models” lists every model your radio has seen. A green [x] means the battery has been assigned to this model. Tapping the model name either assigns or unassigns the battery.

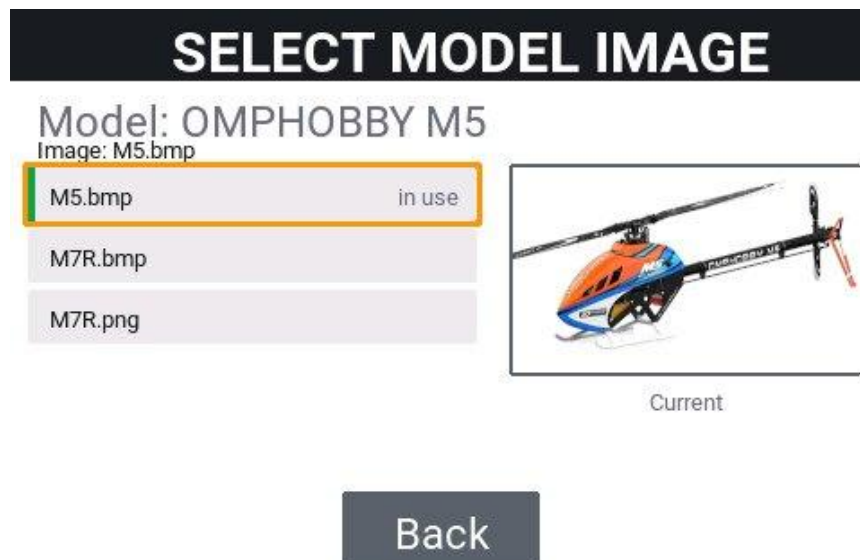


- Num: #n changes which battery profile number this pack uses for that model. Two packs cannot share a number on the same model.
- Delete removes an assignment.

A pack with no assignments at all shows as "Global (all models)" and will appear in the select list for every model.

Models are added to the list automatically the first time they connect, so you usually don't need to type model names by hand.

9. Model images



IMPORTANT:

for the 800x480 version it's recommended to use an image with a resolution of 252x150

For the 480x320 version it's recommended to use an image with a resolution of 192x114

- Put your model images in /WIDGETS/SSODB/Modelimage/
- Tap the photo on the flight screen to open this list. Pick a file, check the preview on the right, and confirm.

10. The post-flight summary

When the link drops, the widget shows a summary of the flight just completed:

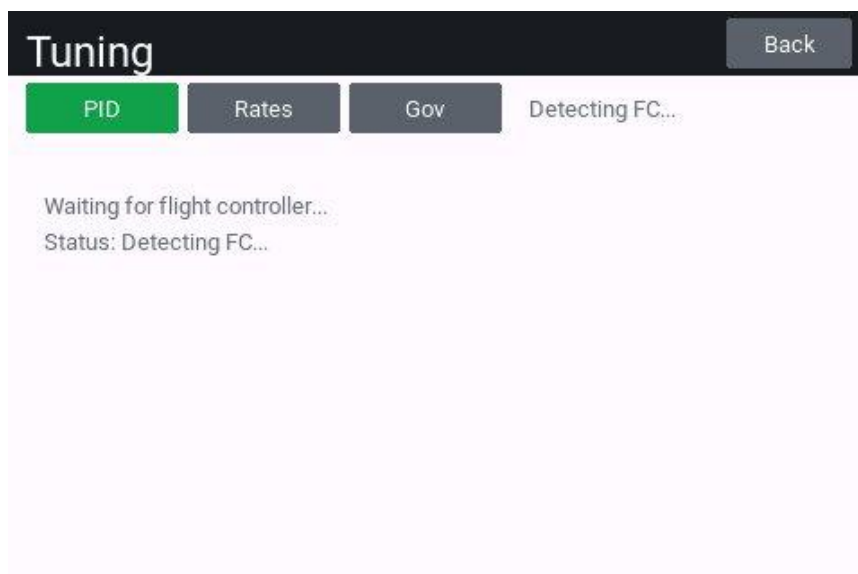
- Model name
- Flight time
- Capacity used
- Maximum ESC temperature
- Maximum current
- Minimum cell voltage
- Worst link quality

Tap Back, press ENTER or press EXIT to dismiss it.

A flight is only counted against the pack once the model has been **armed for at least six seconds**, so a brief arm-and-disarm on the bench does not increment your flight counter.

11. PID tuning

Tap the model name at the top left of the flight screen to open the tuning screen.



Three tabs: PID, Rates and Gov. Values are read live from the flight controller. Use the – and + buttons to change a value and Save to write it back.

Two safety behaviours:

- The screen refuses to open while armed, and beeps instead.
- If you arm while it is open, it closes automatically and returns to the flight screen.

If the tabs show "Waiting for flight controller...", the model is not connected or telemetry is not yet up.

The Rates tab supports ROTORFL-type rates. Other rate types will say so rather than showing values that don't apply.

12. Using the widget without the touchscreen

Every screen can be driven with the rotary encoder and the physical keys, so you can use the widget with the transmitter on a strap.

Control	What it does
Scroll wheel	Moves the highlight from item to item. It wraps around.
ENTER	Activates whatever is highlighted.
EXIT	Goes back one level, or cancels a dialog.

The highlighted item is outlined in orange.

A few screen-specific notes:

- Battery list — the highlight steps through each pack and then its delete button, so you can remove a pack without the touchscreen.
- Delete confirmation — starts on Cancel, so a reflexive ENTER will not delete anything.
- Number entry — the scroll wheel changes the value directly. Capacity moves in steps of 50 mAh, other fields in steps of 1. ENTER confirms.
- Text entry — the highlight steps through the keys. It starts on Done.

13. How your data is stored

Two files in the widget folder hold everything:

- batteries.lua — your packs, their settings and flight counts
- model.lua — which photo belongs to which model

Both are plain text and you can read or back them up from a computer.

They are protected against power loss. The widget writes a verified backup copy (batteries.bak, model.bak) before touching the real file. If you switch the transmitter off in the middle of a save, the widget notices the damaged file next time it starts and recovers from the backup, showing "Batteries restored from backup" on the select screen.

If a save fails outright — a full or write-protected card — you will see "SD write failed" rather than the flight being silently lost.

14. Troubleshooting

Symptom	What to do
The widget shows "Set to fullscreen".	The zone is too small. Set the screen layout to a single full-screen zone.
The engine card always says ENGINE ON.	ModeSrc is not pointing at your engine switch. Fix it in the widget settings.
The headspeed never shows, only ENGINE ON.	Set RpmSrc in the widget settings to your headspeed sensor or channel. Below 100 rpm the card shows ENGINE ON by design — the number only appears once the head is actually turning.
The model name is wrong or says "Unknown".	Give it a few seconds after connecting. If it never resolves, check the Rotorflight Lua scripts are present in /SCRIPTS/RF2/.
My model image doesn't appear.	Check it is in /WIDGETS/SSODB/Modelimage/, is 192 × 114 pixels, and is a supported format. Then reassign it by tapping the photo area.
My battery list came back empty or reset.	The widget falls back to a single "Default" pack if both batteries.lua and batteries.bak are unreadable. Copy a backup back onto the card if you have one.
Flights aren't being counted.	A flight only counts after six seconds armed. Check ArmSrc is set correctly.

Errors shown where "FC sync..." normally appears

Message	Meaning
No mspBatteryConfig	Rotorflight Lua scripts missing or incomplete
No mspApiVersion	As above
API ver timeout	The flight controller did not answer in time
FC sync timeout	The settings write did not complete — reselect the battery
RF2 load failed	/SCRIPTS/RF2/rf2.lua could not be loaded